

Prime Numbers

Chapter-6: Asymmetric Ciphers

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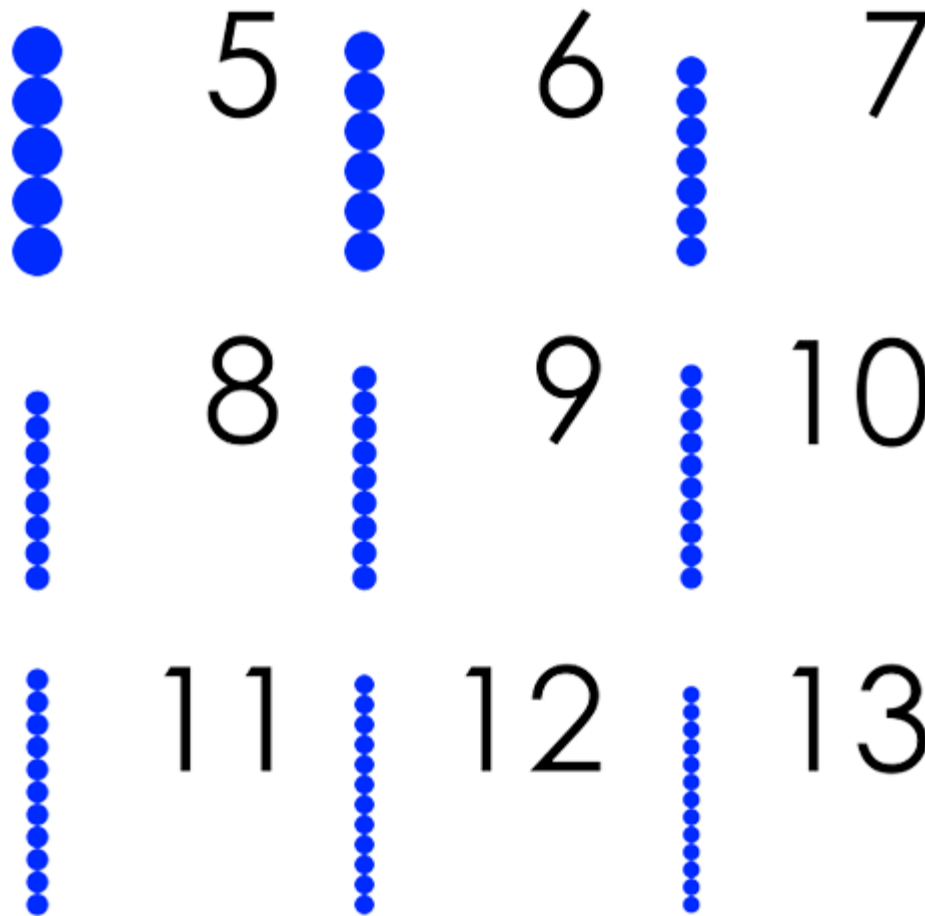
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Prime Numbers

- Prime Numbers are integers greater than 1 with no positive divisors other than 1 and themselves.
- In cryptography, they form the foundation of public-key algorithms, especially RSA.
- **Key property:** It is computationally easy to multiply large primes, but hard to factor the product.
- This asymmetry is central to cryptographic security – known as the integer factorization problem.

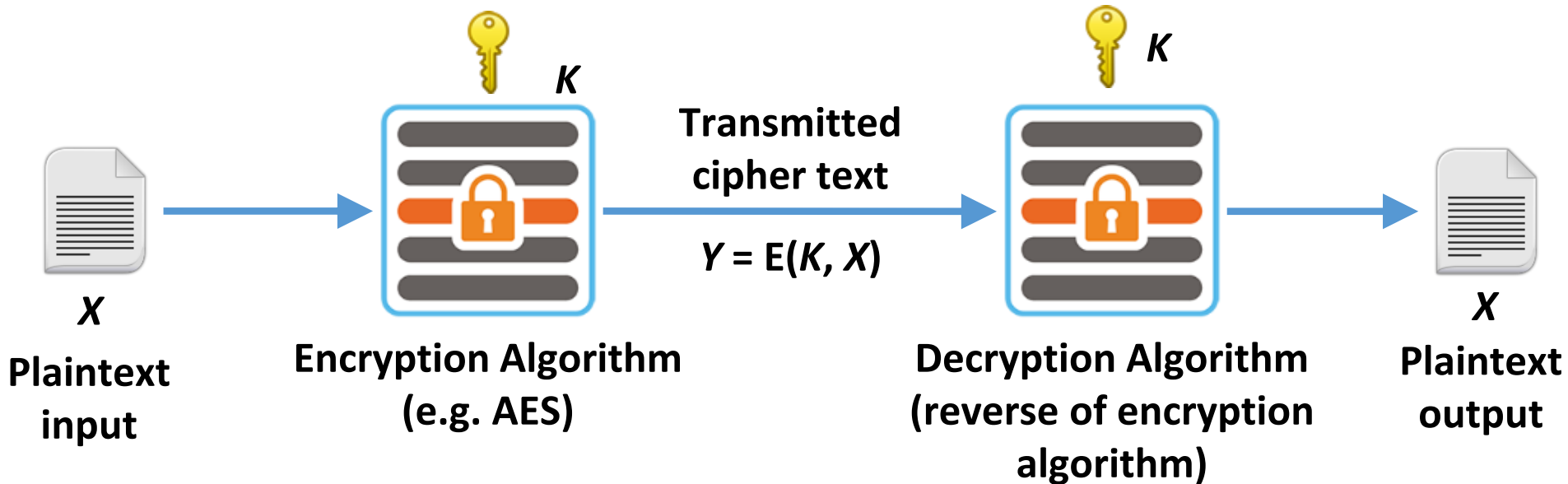
Prime Numbers



Principles of Public-Key Cryptosystems – Symmetric Key Encryption

Secret key shared by
sender and recipient

Secret key shared by
sender and recipient



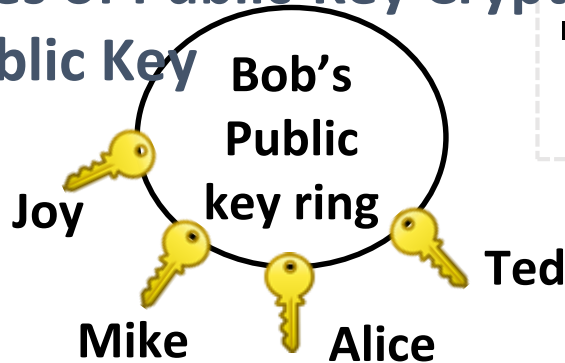
Principles of Public-Key Cryptosystems – Symmetric Key Encryption

- Safe with a strong lock, only Alice and Bob have a copy of the key
- Alice encrypts: locks message in the safe with her key
- Bob decrypts: uses his copy of the key to open the safe



Principles of Public-Key Cryptosystems – Asymmetric Key Encryption with Public Key

- The entire encrypted message serves as a **confidentiality**.



PU_a Alice's public key

PR_a Alice's private key



X



Transmitted cipher text

$$Y = E(PU_a, X)$$



X



Plaintext input

Encryption Algorithm
(e.g. RSA)

Decryption Algorithm

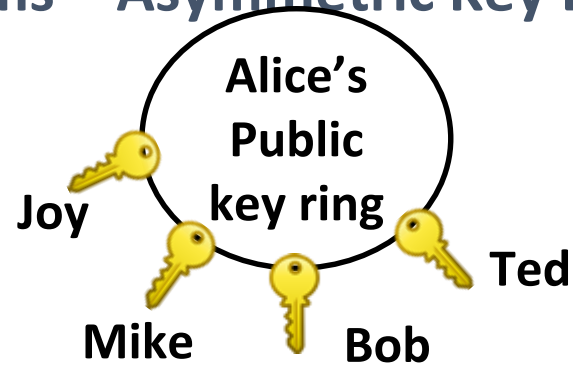
Plaintext output

Bob

Alice

Principles of Public-Key Cryptosystems – Asymmetric Key Encryption with Private Key

- The entire encrypted message serves as a **digital signature**.



PR_b

Bob's private key

PU_b

Bob's public key



X



Transmitted cipher text

$$Y = E(PR_b, X)$$



X



Plaintext input

Encryption Algorithm
(e.g. RSA)

Decryption Algorithm

Plaintext output

Bob

Alice

Multiple Encryption



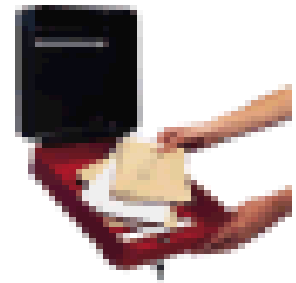
Public Key

New Idea:

Use the good old mailbox principle:

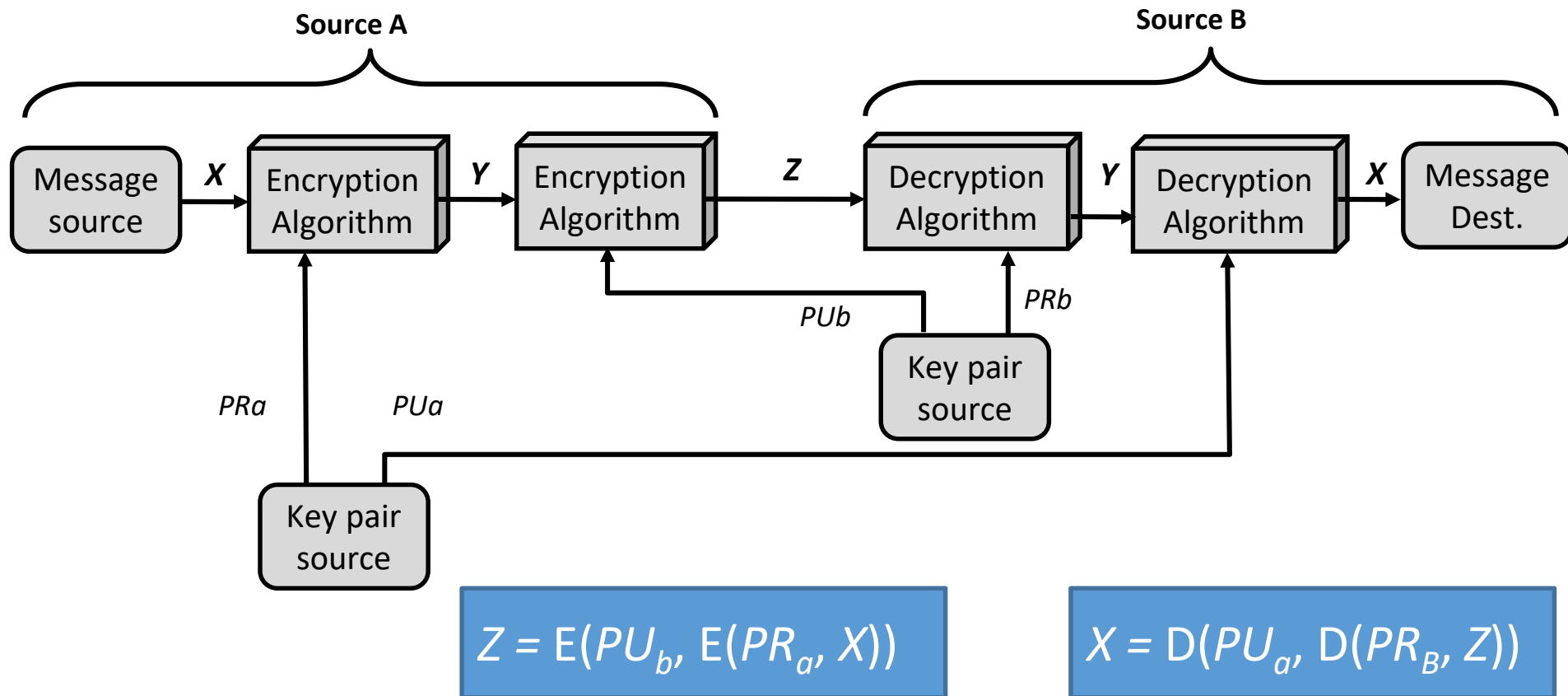
Everyone can drop a letter

But Only the owner has the correct key to open the box



Private Key

Authentication and Confidentiality



Application of Public-Key Cryptosystems

- **Encryption/decryption:** The sender encrypts a message with the recipient's public key.
- **Digital signature:** The sender “signs” a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.
- **Key exchange:** Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

RSA Algorithm

- RSA is a block cipher in which the Plaintext and Ciphertext are represented as integers between 0 and $n-1$ for some value of n .
- Large messages can be broken up into a number of blocks.
- Each block would then be represented by an integer.

- Step-1: Generate Public key and Private key.
- Step-2: Encrypt message using Public key.
- Step-3: Decrypt message using Private key.

RSA Algorithm – Step 1 – Generate Public and Private Keys

- Select two large prime numbers: **p** and **q**
- Calculate modulus : **$n = p * q$**
- Calculate Euler's totient function : **$\phi(n) = (p-1) * (q-1)$**
- Select e such that e is **relatively prime to $\phi(n)$** and **$1 < e < \phi(n)$**

Two numbers are relatively prime if they have no common factors other than 1 or **$\text{gcd}(\phi(n), e) = 1$** .

- Determine **d** such that **$d * e \equiv 1 \pmod{\phi(n)}$**
- Publickey : **$PU = \{ e, n \}$**
- Privatekey : **$PR = \{ d, n \}$**

RSA Algorithm – Step 1 – Generate Public and Private Keys

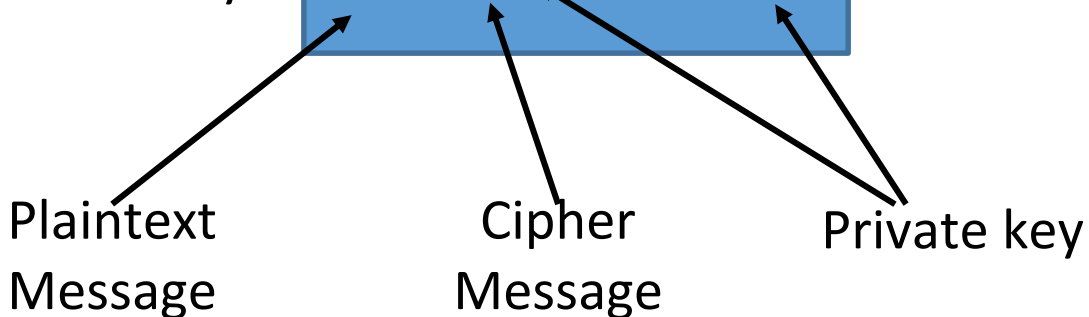
- **Step-2:** Encryption using Public Key:
 - Plaintext: M
 - Ciphertext: $C = M^e \bmod n$
- **Step-3:** Decryption using Private Key:
 - Ciphertext: C
 - Plaintext: $M = C^d \bmod n$

RSA Algorithm – Step 1 – Generate Public and Private Keys

- Select two large prime numbers: $p = 3$ and $q = 11$
 - Calculate modulus : $n = p * q = 3 * 11 = 33$
 - Calculate Euler's totient function : $\phi(n) = (p-1) * (q-1)$
 - $\phi(n) = (3 - 1) * (11 - 1) = 20$
 - Select e such that e is relatively prime to $\phi(n)$ and $1 < e < \phi(n)$
 - We have several choices for e : 3, 7, 11, 13, 17, 19 Let's take $e = 7$
 - Determine d such that $d * e \equiv 1 \pmod{\phi(n)}$
 - $? * 7 \equiv 1 \pmod{20}, 3 * 7 \equiv 1 \pmod{20}$
 - $? * 7 \pmod{20} \equiv 1, 3 * 7 \pmod{20} \equiv 1$
 - Public key : $PU = \{ e, n \}, PU = \{ 7, 33 \}$
 - Private key : $PR = \{ d, n \}, PR = \{ 3, 33 \}$
- This is equivalent to finding d which satisfies $de = 1 + j.\phi(n)$ where j is any integer.
 - We can rewrite this as $d = (1 + j. \phi(n)) / e$

RSA Algorithm – Step 3 - Decryption

- Decryption using Private Key:

$$M = C^d \bmod n$$


Plaintext
Message

Cipher
Message

Private key

$$PR = \{ d, n \}, PR = \{ 3, 33 \}$$

For Ciphertext $C = 20$

$$M = 20^3 \bmod 33$$

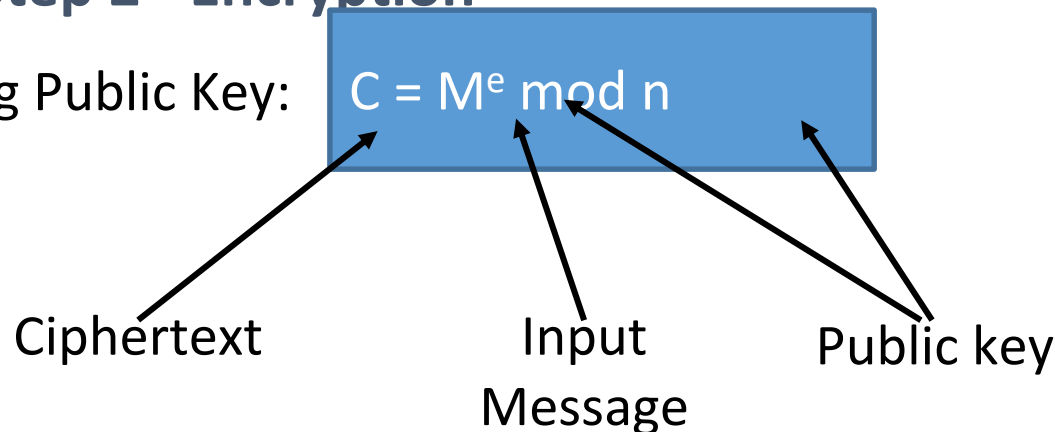
$$M = [(20^1 \bmod 33) \times (20^2 \bmod 33)] \bmod 33$$

$$M = (20 \times 4) \bmod 33 = 80 \bmod 33$$

$$M = 14$$

RSA Algorithm – Step 2 - Encryption

- Encryption using Public Key:



$PU = \{ e, n \}, PU = \{ 7, 33 \}$

For message $M = 14$

$$C = 14^7 \text{ mod } 33$$

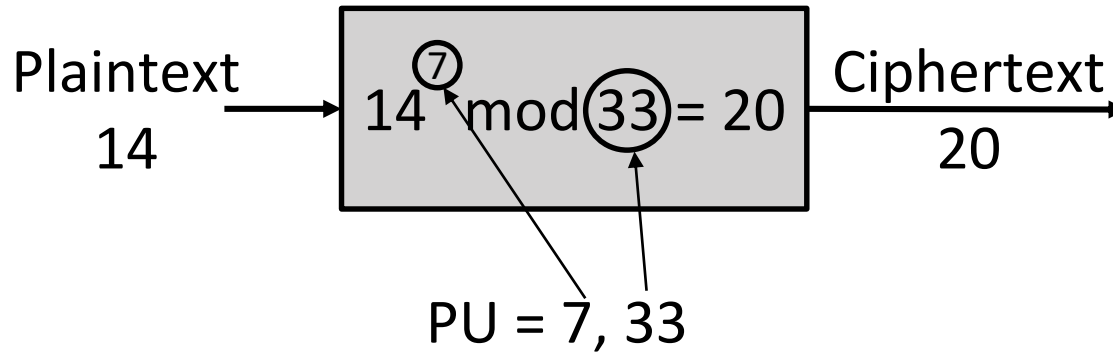
$$C = [(14^1 \text{ mod } 33) \times (14^2 \text{ mod } 33) \times (14^4 \text{ mod } 33)] \text{ mod } 33$$

$$C = (14 \times 31 \times 4) \text{ mod } 33 = 1736 \text{ mod } 33$$

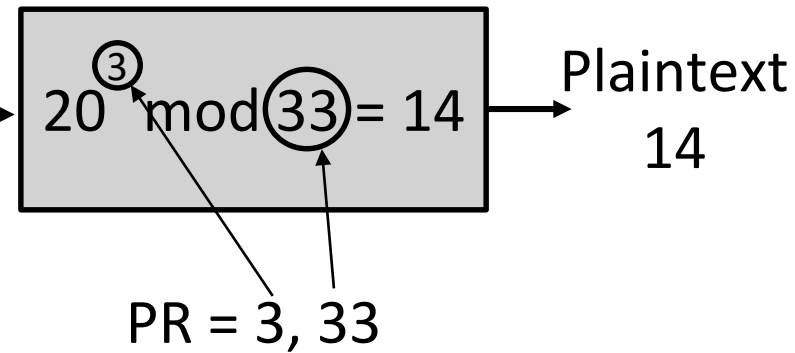
$$C = 20$$

RSA Algorithm

Encryption



Decryption



RSA Algorithm - Example

P and Q are two prime numbers. $P=7$, and $Q=17$. Take public key $E=5$. If plain text value is 10, then what will be cipher text value according to RSA algorithm?

$$n = 119$$

$$\phi(n) = 96$$

$$e = 5$$

$$d = 77$$

$$PU = \{ 5, 119 \}$$

$$PR = \{ 77, 119 \}$$

$$C = 105 \bmod 119 \Rightarrow C = 40$$

RSA Algorithm – Step 1 – Generate Public and Private Keys

- P and Q are two prime numbers. $P=17$, and $Q=11$. If plain text value is 88, then what will be cipher text value according to RSA algorithm?
- $n = 187$
- $\phi(n) = 160$
- $e = 7$
- $d = 23$
- $PU = \{ 7, 187 \}$
- $PR = \{23, 187\}$
- $C = 887 \bmod 187 \Rightarrow C = 11$

Diffie Hellman Key Exchange

The purpose of the **Diffie-Hellman algorithm** is to enable two users to securely exchange a key that can be used for subsequent encryption of message.

The effectiveness of an algorithm is depends on the difficulty of computing **discrete logarithms**.

Diffie Hellman Key Exchange

- Let p be a prime number.
- Then a is a primitive root for p , if the powers of a modulo p generates all integers from 1 to $p - 1$ in some permutation.

$$a \bmod p, a^2 \bmod p, \dots, a^{p-1} \bmod p$$

- Example: $p = 7$ then primitive root is 3 because powers of 3 mod 7 generates all the integers from 1 to 6.

$$3^1 = 3 \equiv 3 \pmod{7}$$

$$3^2 = 9 \equiv 2 \pmod{7}$$

$$3^3 = 27 \equiv 6 \pmod{7}$$

$$3^4 = 81 \equiv 4 \pmod{7}$$

$$3^5 = 243 \equiv 5 \pmod{7}$$

$$3^6 = 729 \equiv 1 \pmod{7}$$

Diffie Hellman Key Exchange

For any integer ***b*** and a primitive root ***a*** of prime number ***p***, we can find a unique exponent ***i*** such that

$$b = a^i \pmod{p} \text{ where } 0 \leq i \leq (p - 1)$$

The exponent ***i*** is referred as the discrete logarithm of ***b*** for the base ***a***, mod ***p***. It expressed as below.

$$i = \text{dlog}_{a,p}(b)$$

Diffie Hellman Key Exchange

User A and User B agree on two large prime numbers q and α . User A and User B can use insecure channel to agree on them.

User A selects a random integer $X_A < q$ and calculates Y_A .

User B selects a random integer $X_B < q$ and calculates Y_B .

Diffie Hellman Key Exchange

Global Public Elements

q

prime number

α

$\alpha < q$ and α is primitive root of q

User A Key Generation

Select private X_A

$$X_A < q$$

Calculate public Y_A

$$Y_A = \alpha^{X_A} \bmod q$$

User B Key Generation

Select private X_B

$$X_B < q$$

Calculate public Y_B

$$Y_B = \alpha^{X_B} \bmod q$$

Diffie Hellman Key Exchange

User A Key Generation

Select private X_A

$$X_A < q$$

Calculate public Y_A

$$Y_A = \alpha^{X_A} \bmod q$$

User B Key Generation

Select private X_B

$$X_B < q$$

Calculate public Y_B

$$Y_B = \alpha^{X_B} \bmod q$$

Calculation of Secret Key by User A

$$K = (Y_B)^{X_A} \bmod q$$

Calculation of Secret Key by User b

$$K = (Y_A)^{X_B} \bmod q$$

Diffie Hellman Key Exchange

User A Key Generation

Private X_A $X_A < q$,

Public Y_A

$$Y_A = \alpha^{X_A} \bmod q$$

User B Key Generation

Private X_B $X_B < q$,

Public Y_B

$$Y_B = \alpha^{X_B} \bmod q$$

Secret Key by User A : $K = (Y_B)^{X_A} \bmod q$

Secret Key by User B : $K = (Y_A)^{X_B} \bmod q$

$$\begin{aligned} K_A &= (Y_B)^{X_A} \bmod q \\ K &= (\alpha^{X_B} \bmod q)^{X_A} \bmod q \\ K &= (\alpha^{X_B})^{X_A} \bmod q \\ K &= \alpha^{X_B X_A} \bmod q \\ K &= (\alpha^{X_A})^{X_B} \bmod q \\ K &= (\alpha^{X_A} \bmod q)^{X_B} \bmod q \\ K_B &= (Y_A)^{X_B} \bmod q \end{aligned}$$

Diffie Hellman Key Exchange

Alice

Alice and Bob share a prime number q and an integer a , such that $a < q$ and a is a primitive root of q

Alice generates a private key X_A such that $X_A < q$

Alice calculates a public key
 $Y_A = a^{X_A} \bmod q$

Alice receives Bob's public key Y_B in plaintext

Alice calculates shared secret key
 $K = (Y_B)^{X_A} \bmod q$

Bob

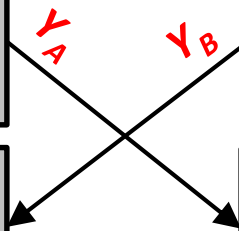
Alice and Bob share a prime number q and an integer a , such that $a < q$ and a is a primitive root of q

Bob generates a private key X_B such that $X_B < q$

Bob calculates a public key
 $Y_B = a^{X_B} \bmod q$

Bob receives Alice's public key Y_A in plaintext

Bob calculates shared secret key
 $K = (Y_A)^{X_B} \bmod q$



Diffie Hellman Key Exchange

Alice and bob agrees on a prime number $q = 23$

$\alpha = 5$ as primitive root of q .

Alice selects a private integer $X_A = 6$

Alice computes $Y_A = \alpha^{X_A} \bmod q \Rightarrow Y_A = 5^6 \bmod 23 = 8$

Bob selects a private integer $X_B = 15$

Bob computes $Y_B = \alpha^{X_B} \bmod q \Rightarrow Y_B = 5^{15} \bmod 23 = 19$

Alice sends Y_A to Bob and Bob sends Y_B to Alice.

Alice computes key $K = (Y_B)^{X_A} \bmod q \Rightarrow K = (19)^6 \bmod 23$

$$K = 2$$

Bob computes key $K = (Y_A)^{X_B} \bmod q \Rightarrow K = (8)^{15} \bmod 23$

$$K = 2$$

Diffie Hellman Key Exchange

Alice and bob agrees on a prime number $q = 19$

$\alpha = 10$ as primitive root of q .

Alice selects a private integer $X_A = 7$

Alice computes $Y_A = \alpha^{X_A} \bmod q \Rightarrow Y_A = 10^7 \bmod 19 = 15$

Bob selects a private integer $X_B = 8$

Bob computes $Y_B = \alpha^{X_B} \bmod q \Rightarrow Y_B = 10^8 \bmod 19 = 17$

Alice sends Y_A to Bob and Bob sends Y_B to Alice.

Alice computes key $K = (Y_B)^{X_A} \bmod q \Rightarrow K = (17)^7 \bmod 19$
 $K = 5$

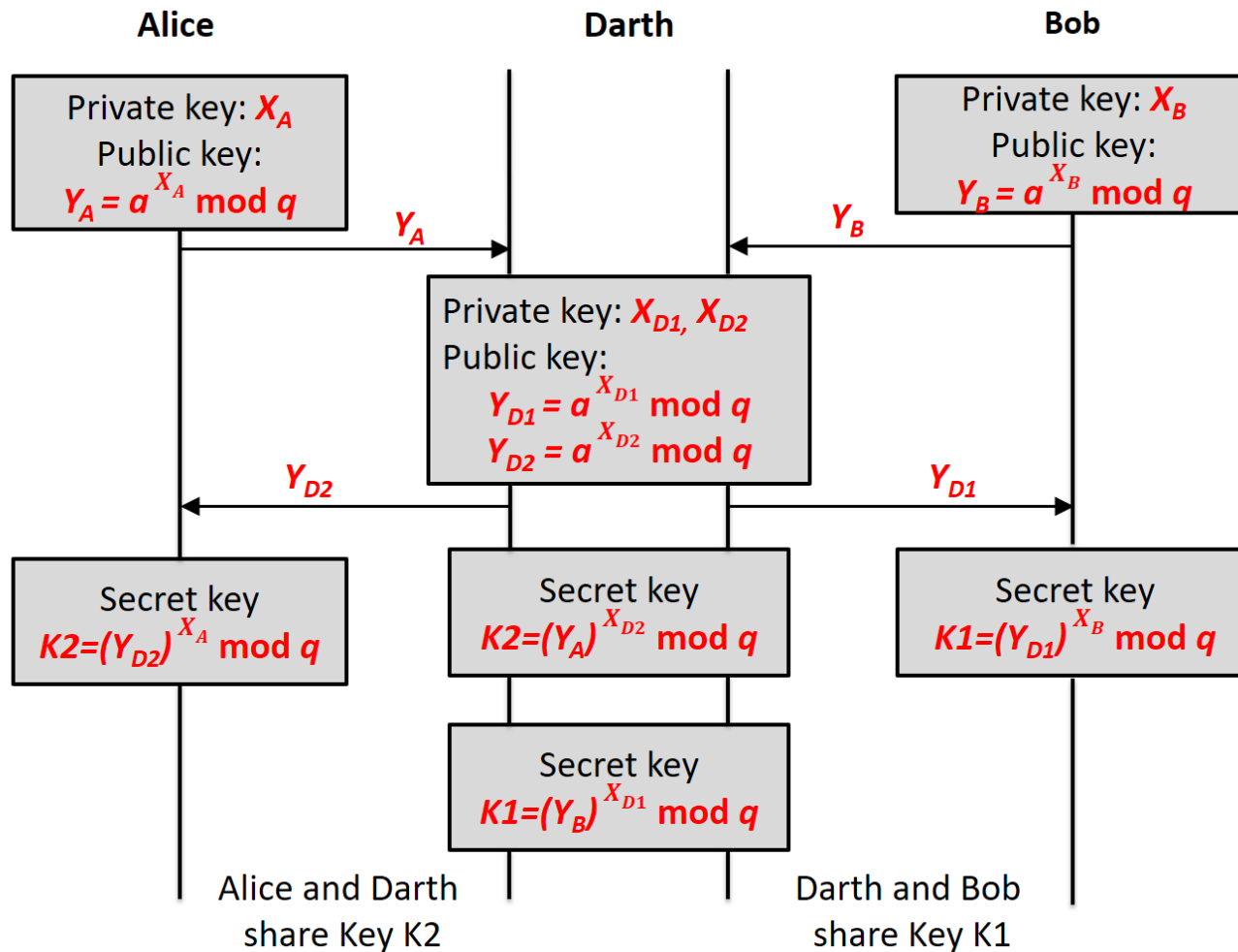
Bob computes key $K = (Y_A)^{X_B} \bmod q \Rightarrow K = (15)^8 \bmod 19$
 $K = 5$

Man in the Middle

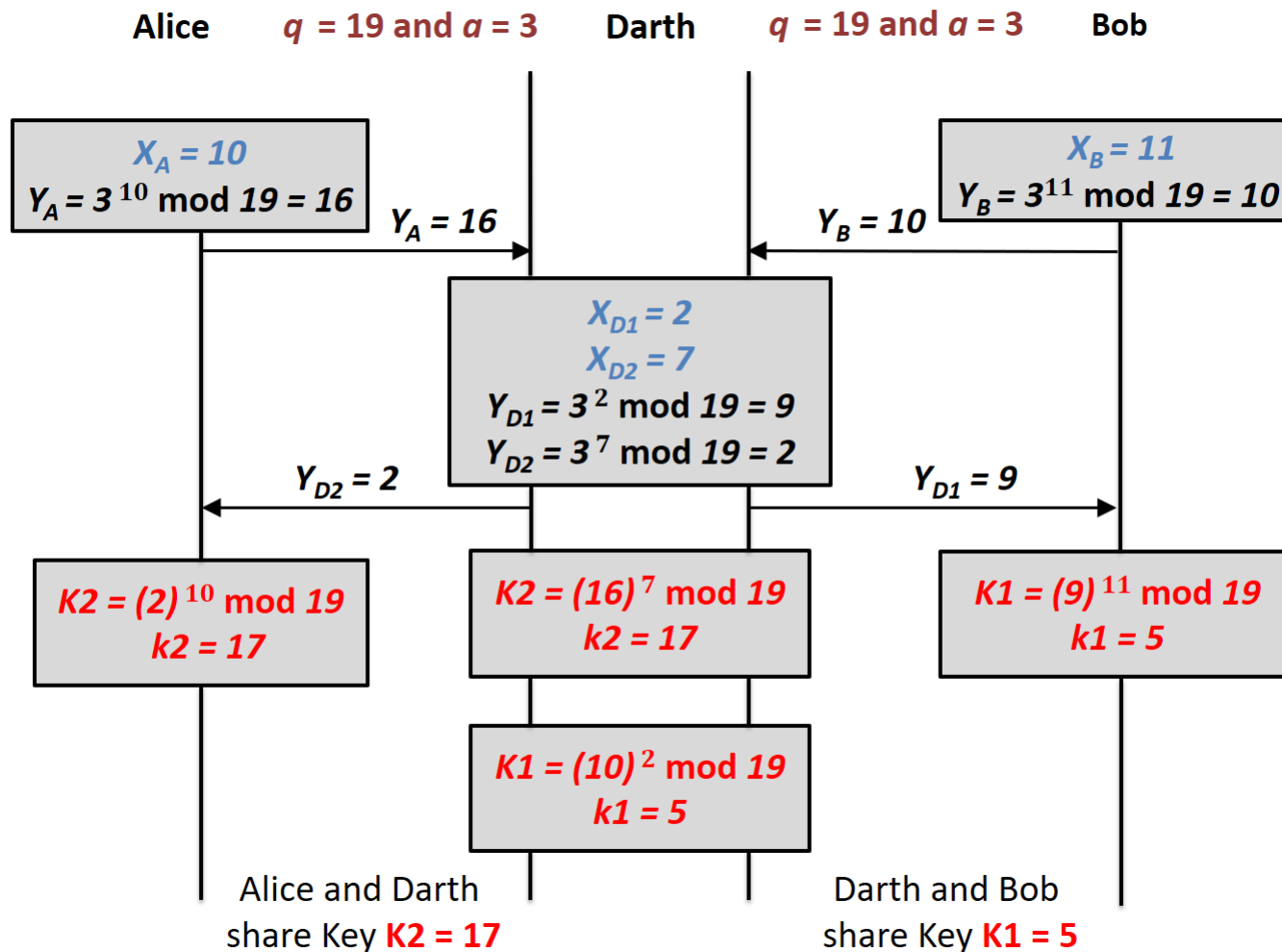
Suppose Alice and Bob wish to exchange keys, and Darth is the adversary.

1. Darth prepares for the attack by generating two random private keys X_{D1} and X_{D2} and then computes corresponding public keys Y_{D1} and Y_{D2} .
2. Alice transmits Y_A to Bob.
3. Darth intercepts Y_A and transmits Y_{D1} to Bob. Darth also calculates $K_2 = (Y_A)^{X_{D2}} \bmod q$.
4. Bob receives Y_{D1} and calculates $K_1 = (Y_{D1})^{X_B} \bmod q$.
5. Bob transmits Y_B to Alice.
6. Darth intercepts Y_B and transmits Y_{D2} to Alice. Darth calculates $K_1 = (Y_B)^{X_{D1}} \bmod q$.
7. Alice receives Y_{D2} and calculates $K_2 = (Y_{D2})^{X_A} \bmod q$.

Man in the Middle



Man in the Middle



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